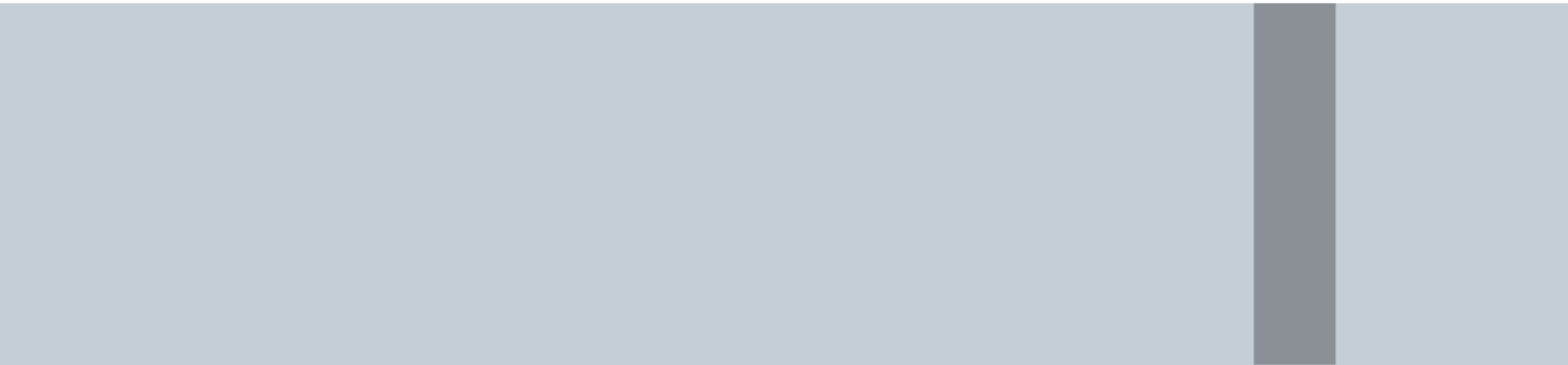
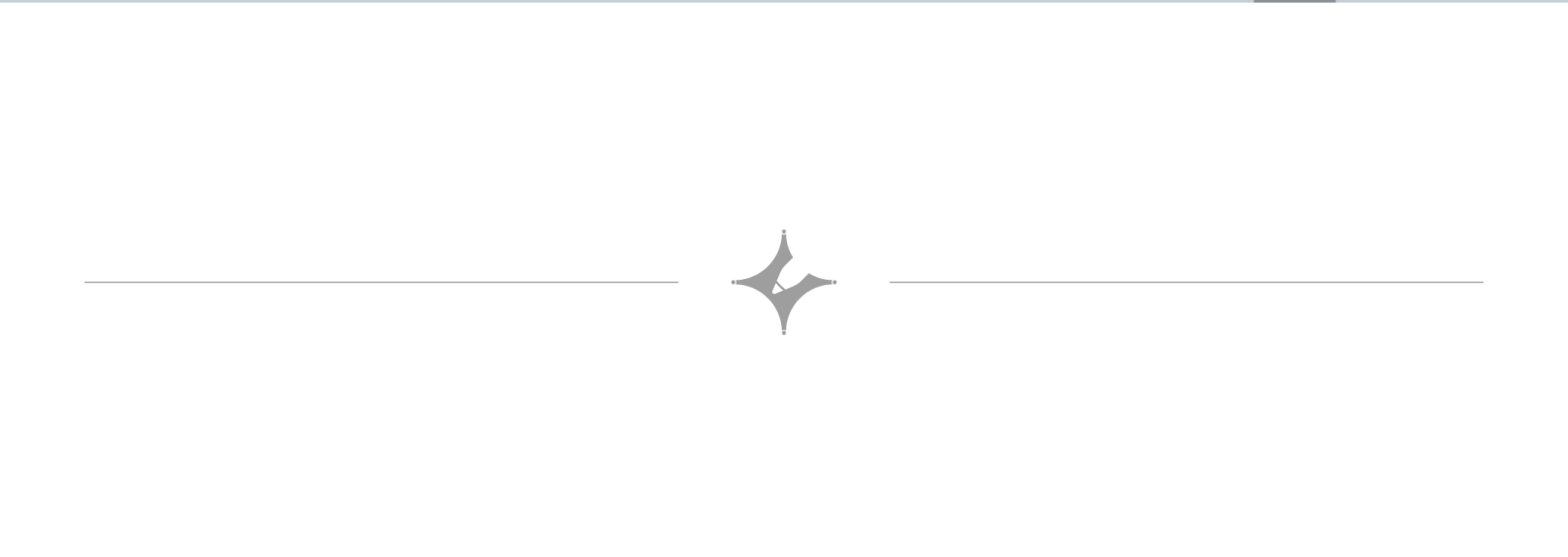
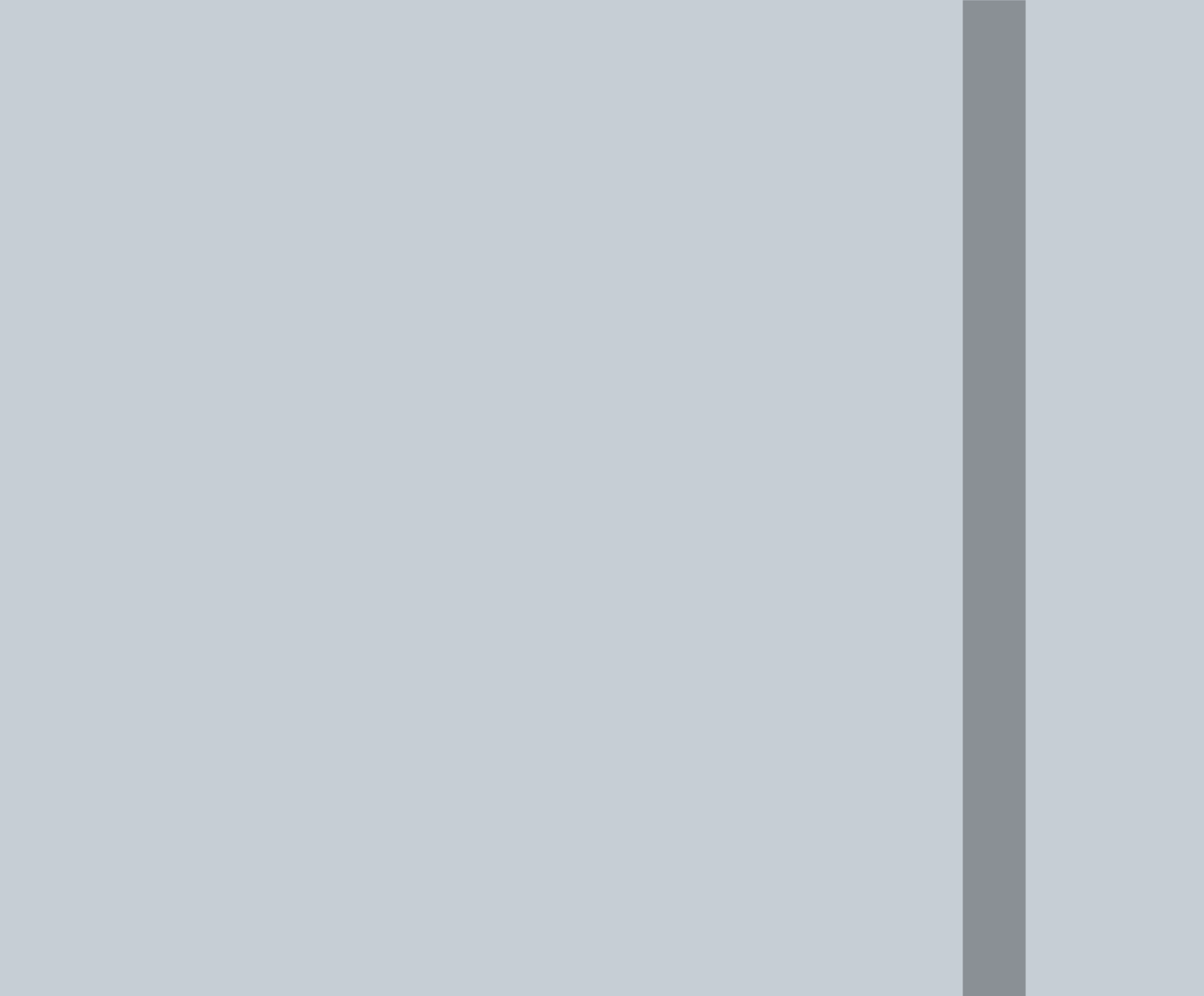




Time, Speed and distance

CONCEPT

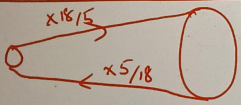
$$m/s \xrightarrow{\times 18/5} km/hr$$

$$km/hr \xleftarrow{\times 5/18} m/s$$

$$S = \frac{D}{T} \quad T = \frac{D}{S} \quad D = S \times T$$

$$S_{avg} = \frac{\text{Total } D}{\text{Total } T} = \frac{(D_1 + D_2)}{(T_1 + T_2)}$$

$$= \frac{2S_1 S_2}{S_1 + S_2} \quad [D_1 = D_2]$$



9. Flash and Quicksilver are 90 km away from one another. They are starting to move towards each other simultaneously, Flash at a speed of 10 kmph and Quicksilver at a speed of 5 kmph. If after every hour they double their speeds, what is the distance that Flash will travel until he meets Quicksilver?

A) 45 km ☒ B) 60 km ☐ C) 30 km ☐ D) 80 km

$$\rightarrow D = S \times T$$

$$\Rightarrow D \propto S \quad (10:5)$$

$\rightarrow$ If Speed is 2:1 then distance travelled will also be in 2:1 ratio. So, 90 = 60:30

10. Without stoppages a train travels a certain distance with a speed of 50 km/h and with stoppages it covers the same distance with an average speed of 40 km/h. How many minutes per hour does the train stop?

A) 10 min/h ☐ B) 15 min/h ☐ C) 12 min/h ☒ D) 8 min/h

$$\frac{10}{50} \times 60 = 12$$

w/o stop	1 hr	50 km	(10 km)
with stop	1 hr	40 km	

$$T = \frac{10}{50} = \frac{1}{5}$$

$$\frac{1}{5} \times 60 = 12 \text{ min/hr}$$

(short cut).

$$\frac{\text{change in something}}{\text{original}} \times 100\%$$

$$\Rightarrow \frac{50 - 40}{50} \times 60$$

$$\Rightarrow \frac{10}{50} \times 60$$

(logic).

(100% of anything which in this case is 60 mins.

11. A car travels from A to B at V_1 km/hr, travels back from B to A at V_2 km/hr. The average speed of the car is:

A) $2V_1 V_2 / (V_1 + V_2)$

B) $2V_1 V_2 / (2V_1 + V_2)$

C) $3V_1 V_2 / (2V_1 + V_2)$

D) $3V_1 V_2 / (V_1 + 2V_2)$

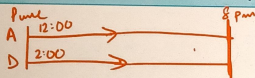
	1	2	3
D	D	D	D
S	V_1	V_2	V_2
T	$\frac{D}{V_1}$	$\frac{D}{V_2}$	$\frac{D}{V_2}$

$$S_{avg} = \frac{3D}{\frac{D}{V_1} + \frac{2D}{V_2}} = \frac{3}{\frac{1}{V_1} + \frac{2}{V_2}}$$

$$= \frac{3V_1 V_2}{2V_1 + V_2}$$

15. The Alpha Express left Pune for Mumbai at noon sharp. Two hours later, the Deccan Queen started from Pune in the same direction. The Queen overtook the Alpha Express at 8 p.m. Find the average speed of the two trains over the journey if the sum of their average speeds is 70 kmph.

A) 35 kmph ☒ B) 34.28 kmph ☐ C) 50 kmph ☐ D) 12 kmph



$$D_A = D_D$$

$$S_A T_A = S_D T_D$$

$$x \cdot 8 = (70 - x) \cdot 6$$

$$4x = 3 \times 70 - 3x$$

$$7x = 3 \times 70$$

$$x = 30$$

$$S_A + S_D = 70$$

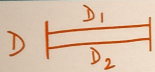
$$S_A = 30$$

$$S_D = 40$$

$$S_{avg} = \frac{2 \times 30 \times 40}{30 + 40} = 34.28$$

16. Two trains for Mumbai leave Delhi at 6 a.m. and 6:45 a.m. and travel at 100 kmph and 136 kmph respectively. How many kilometers from Delhi will the two trains be together?

- A) 272 km B) 260 km C) 283.33 km D) 262.4 km



$$S_1 T_1 = S_2 T_2$$

$$100 \times T = 136 \left(T - \frac{45}{60}\right)$$

$$100T = 136T - 136 \times \frac{3}{4}$$

$$36T = 136 \times \frac{3}{4}$$

1M

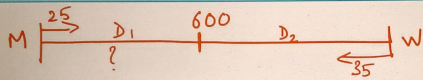
$$T = \frac{136 \times \frac{3}{4}}{136 - 100} = \frac{17}{6} \text{ hrs}$$

$$D = 100 \times \frac{17}{6} = 283.33$$

Both trains meet at the same time & 2nd train started 45 mins late meaning it took 45 mins less time for this train to reach the place.

17. Mirzapur and Wasseypur are two stations 600km apart. A train starts from Mirzapur and moves towards Wasseypur at the rate of 25km/h. After 2 hours, another train starts from Wasseypur in the opposite direction at the rate of 35km/h. How far from Mirzapur will they cross each other?

- A) 250 km B) 300 km C) 279.166 km D) 475 km



$$D_1 + D_2 = 600$$

$$25 \times T + 35(T - 2) = 600$$

$$25T + 35T - 70 = 600$$

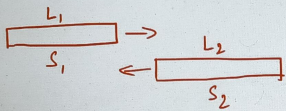
$$60T = 670$$

$$T = \frac{67}{6}$$

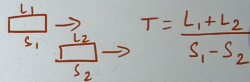
$$D_1 = 25 \times \frac{67}{6} = 279.166$$

2nd train starts 2 hr late.

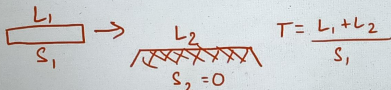
CONCEPT



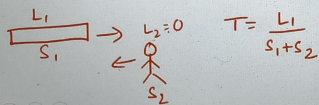
$$T = \frac{L_1 + L_2}{S_1 + S_2}$$



$$T = \frac{L_1 + L_2}{S_1 - S_2}$$



$$T = \frac{L_1 + L_2}{S_1}$$



$$T = \frac{L_1}{S_1 + S_2}$$

few more concepts